

Reconstructing frustrated signs from amplitudes with fixed-node and Krylov updates

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We split a real wave function into signs and positive amplitudes, $\psi = sa$, and update them alternately in the frustrated square-lattice J_1 - J_2 model at $J_2/J_1 = 0.5$. The signs follow from a label-free Krylov step, $s' = s \operatorname{sgn}[T - (H\psi)/\psi]$; the amplitude from lattice fixed node, which cannot raise the energy. Given the exact amplitude, three to four Krylov steps recover the exact ground-state signs on a 4×4 cluster for $0.4 \leq J_2/J_1 \leq 1$. Alternating both updates with exact fixed node converges to the exact ground state on 4×4 and 20-site clusters, about three times faster with Anderson acceleration. On 6×6 and 8×8 , Krylov signs built from a neural-network amplitude reduce the fixed-node energy gap to the network from 4×10^{-3} to 6×10^{-4} per site, without closing it. Iterating further requires learning the fixed-node amplitude from walkers, which remains the open problem.

I. INTRODUCTION

The Monte Carlo sign problem is severe in frustrated spin systems [1, 2]. We use the square-lattice J_1 - J_2 Heisenberg model as a benchmark. At $J_2 = 0$ its ground-state signs obey the Marshall rule [3]; near $J_2/J_1 = 1/2$ that rule fails and accurate variational states become nontrivial [4, 5].

We ask whether signs and positive amplitudes can be improved separately. For a real wave function we write

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \quad a(x) = |\psi(x)| > 0. \quad (1)$$

Given a , a single Krylov step updates the signs. Given s , lattice fixed node gives a sign-free positive-amplitude problem [6]. Section II defines both updates. Section III tests them in order of increasing difficulty: first the sign update alone with an exact amplitude, then the exact feedback loop, then larger lattices, and finally the open problem of learning the amplitude from fixed-node walkers.

II. METHODS

A. Model and error measures

We study

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (2)$$

on periodic lattices with $J_1 = 1$ in the $S^z = 0$ sector. For a spin configuration x , the Marshall rule is $s_M(x) = (-1)^{N_\downarrow^A(x)}$ up to an overall sign, where $N_\downarrow^A$ counts down spins on one checkerboard sublattice.

When the exact ground state $\psi_0 = s_0 a_0$ is available, we fix the irrelevant overall sign and measure the sign error by the wrong-sign probability

$$w_s = \sum_{x: s(x) \neq s_0(x)} a_0(x)^2, \quad (3)$$

and the energy error by $\epsilon_{\text{rel}} = (E - E_0)/|E_0|$.

B. One-step Krylov sign update

For $\psi = sa$, define the local energy

$$r_s(x) = \frac{(H\psi)(x)}{\psi(x)}. \quad (4)$$

Because $[(T - H)\psi](x) = [T - r_s(x)]\psi(x)$, the sign of the shifted Krylov vector is

$$s_K(x) = s(x) \operatorname{sgn}[T - r_s(x)] : \quad (5)$$

the sign of every configuration with $r_s(x) > T$ is flipped. Only this sign is kept; the amplitude factor $|T - r_s(x)|$ is discarded, so the method is not repeated Lanczos multiplication. The threshold T never uses the target signs. For exact systems, T minimizes the variational energy of $a(x)s_K(x)$ over all thresholds. For sampled systems, T is obtained from a two-cluster split of the sampled $r_s(x)$ values.

C. Fixed-node amplitude update

For a trial wave function ψ_T , an off-diagonal matrix element connecting x and y is sign violating when $H_{xy}\psi_T(y)/\psi_T(x) > 0$. Lattice fixed node removes such terms and adds their contribution to the diagonal [6]. The resulting Hamiltonian is sign free, keeps the signs of ψ_T fixed, and its ground-state energy E_{FN} is a variational upper bound. Let $\phi_{\text{FN}}(x) > 0$ be its positive ground-state amplitude. In an exact small-system calculation, ϕ_{FN} replaces a directly before the next application of Eq. (5).

D. Accelerating the amplitude iteration

With exact fixed node, the loop is a fixed-point iteration for the amplitude, $x_{k+1} = g(x_k)$ with $x = \log a$ and $g(x_k) = \log \phi_{\text{FN}}[a_k, s_k]$. We accelerate it with Anderson mixing [7, 8]: with the residuals $f_k = g(x_k) - x_k$ and the differences $\Delta f_j, \Delta g_j$ of the last m iterates,

$$x_{k+1} = g(x_k) - \sum_j \gamma_j \Delta g_j, \quad \gamma = \arg \min_{\gamma} \left\| f_k - \sum_j \gamma_j \Delta f_j \right\|. \quad (6)$$

We use $m = 3$ and clear the history whenever the fixed-node energy rises or a step becomes too large. The sign step is unchanged.

E. Learning the amplitude from walkers

In a walker calculation ϕ_{FN} is not available as a function. The stationary walker distribution is instead the mixed distribution

$$f(x) \propto a(x)\phi_{\text{FN}}(x). \quad (7)$$

If a positive model is trained by maximum likelihood with $p_{\theta}(x) \propto a_{\theta}(x)^2$, the exact maximum-likelihood target is

$$a_{\text{new}}^2(x) \propto a(x)\phi_{\text{FN}}(x), \quad (8)$$

the geometric mean of the current and fixed-node amplitudes. We precondition the likelihood gradient g with stochastic reconfiguration (SR),

$$\delta\theta = (S + \lambda I)^{-1}g, \quad (9)$$

where S is the quantum geometric tensor of $\log a_{\theta}$.

F. Numerical setup

Exact calculations use two periodic clusters (Fig. 1): the 4×4 square ($N = 16$, 12 870 states with $S^z = 0$) and a tilted 20-site torus with periodic vectors $(4, 0)$ and $(1, 5)$.

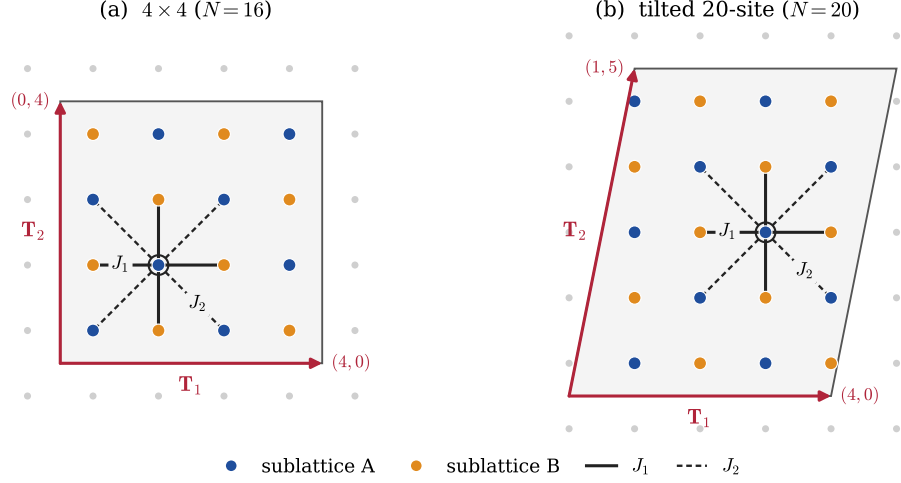


FIG. 1. Exact clusters with periodic boundaries: (a) 4×4 , (b) tilted 20-site torus. Colours: Marshall sublattices; arrows: periodic vectors; J_1/J_2 bonds shown for one site.

For 6×6 and 8×8 we use a previously optimized translationally invariant vision-transformer neural quantum state (ViT) at $J_2/J_1 = 0.5$. Fixed-node energies are mixed estimates from walker populations of size M and projection length $\beta = 1.2$; 6×6 uses $M = 32$ and $M = 128$, 8×8 uses $M = 128$. Statistical conventions are summarized in Appendix B.

III. RESULTS

A. Krylov sign update

We test the sign update alone, holding the amplitude fixed, with amplitudes of decreasing quality: exact, approximate, and a neural network.

1. Exact amplitude

The first benchmark uses the exact 4×4 cluster:

1. **Guide.** Take the exact ground-state amplitude $a_0(x) = |\psi_0(x)|$ from exact diagonalization and give it the Marshall sign, $\psi(x) = s_M(x) a_0(x)$. The amplitude is exact, so all error is in the signs. (For $J_2/J_1 \geq 0.8$ the stripe sign replaces Marshall.)
2. **One Krylov sign step.** Compute $r(x) = (H\psi)(x)/\psi(x)$, choose the threshold T that minimizes the energy of $a_0 s'$, and set $s'(x) = s_M(x) \text{sgn}[T - r(x)]$, Eq. (5). The amplitude a_0 is not changed.
3. **Compare with exact diagonalization.** Measure the wrong-sign probability w_s , Eq. (3), and the relative energy error ϵ_{rel} of $\psi = s_M a_0$ (before) and $\psi' = s' a_0$ (after).
4. **Repeat.** Apply the Krylov sign step again to $s' a_0$, with a new threshold computed from the new local energies, still keeping a_0 fixed and never using the exact signs.

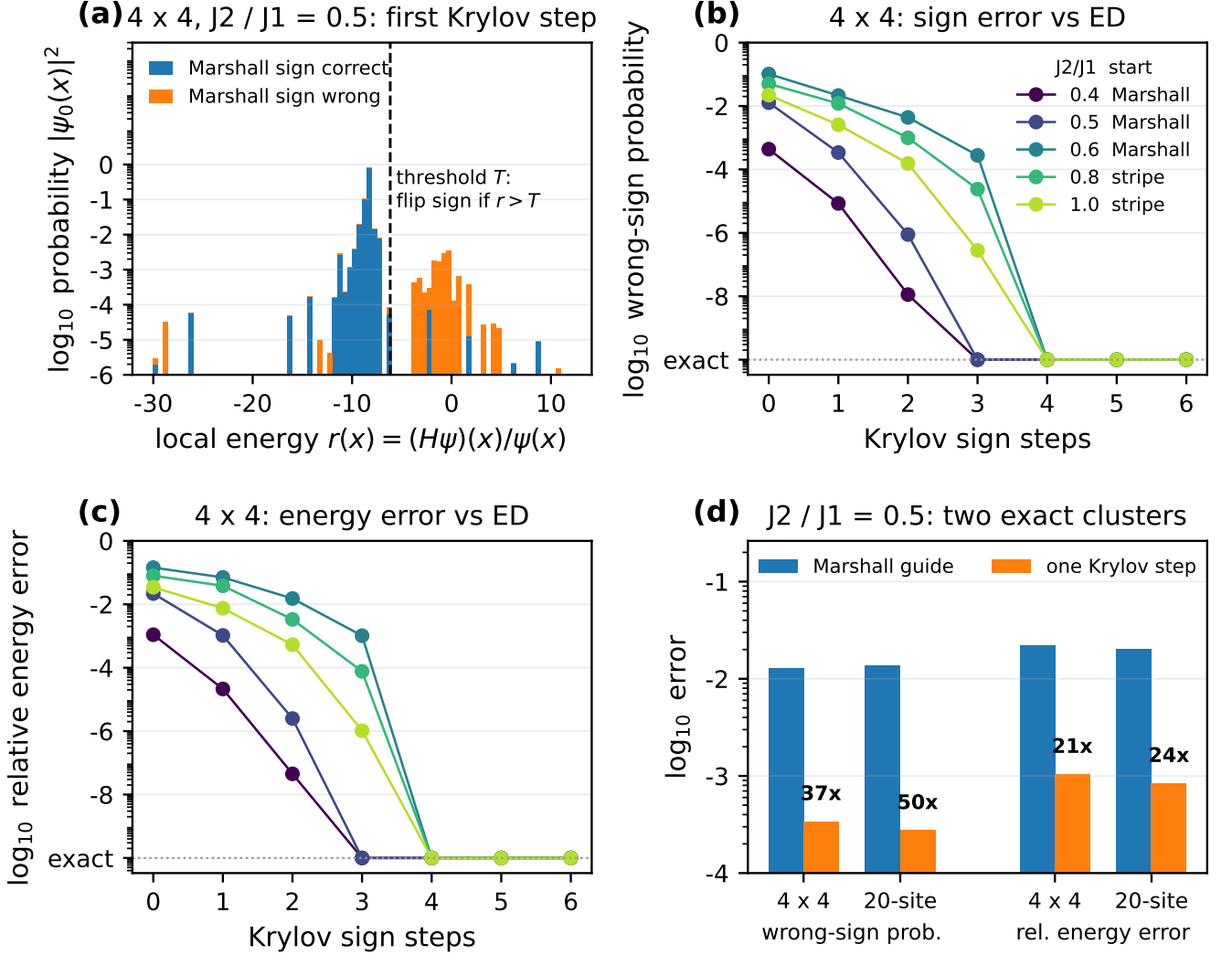


FIG. 2. Sign correction by Krylov steps, starting from Marshall signs. Only the signs change; the amplitude stays the exact ED one. (a) $r(x)$ at $J_2/J_1 = 0.5$; dashed line: threshold T . (b) Wrong-sign probability and (c) energy error vs number of Krylov sign steps, compared with ED. (d) First step on the 4×4 and 20-site clusters (Fig. 1).

Figure 2(a) shows why one step works at $J_2/J_1 = 0.5$. The configurations on which the Marshall sign is wrong have systematically larger local energies $r(x)$ than the correctly signed ones, and the threshold T separates the two groups: 98% of the wrong-sign probability lies at $r > T$. One step lowers the wrong-sign probability from 1.27×10^{-2} to 3.41×10^{-4} , a factor 37, and the relative energy error from 2.19×10^{-2} to 1.04×10^{-3} , a factor 21. The same first step gives factors 50 and 24 on the independent 20-site cluster [Fig. 2(d)].

One step is close to the ground state but not exact: the fidelity with ψ_0 is 0.9986. Repeating the step removes the remaining errors [Fig. 2(b,c)]. At $J_2/J_1 = 0.5$ the second step gives $w_s = 8.9 \times 10^{-7}$ and $\epsilon_{\text{rel}} = 2.5 \times 10^{-6}$, and the third step reaches the exact ground-state signs. For every coupling from $J_2/J_1 = 0.4$ to 1.0, including the most frustrated point 0.6 where the Marshall guide has 10% wrong-sign probability, at most four steps reach the exact signs. With an exact amplitude, the sign problem on this cluster is therefore solved by a few label-free Krylov steps. The question is what happens when the amplitude is not exact.

2. Approximate amplitude

Next the amplitude is only approximate. Keeping the target $J_2/J_1 = 0.5$, we use as fixed amplitude the exact ground-state amplitude of a different coupling, $a(x) = |\psi_0^{J'_2}(x)|$ with $J'_2 = 0-0.6$, which gives a physical, structured

amplitude error. The guide again has Marshall signs, and the Krylov steps are repeated as before.

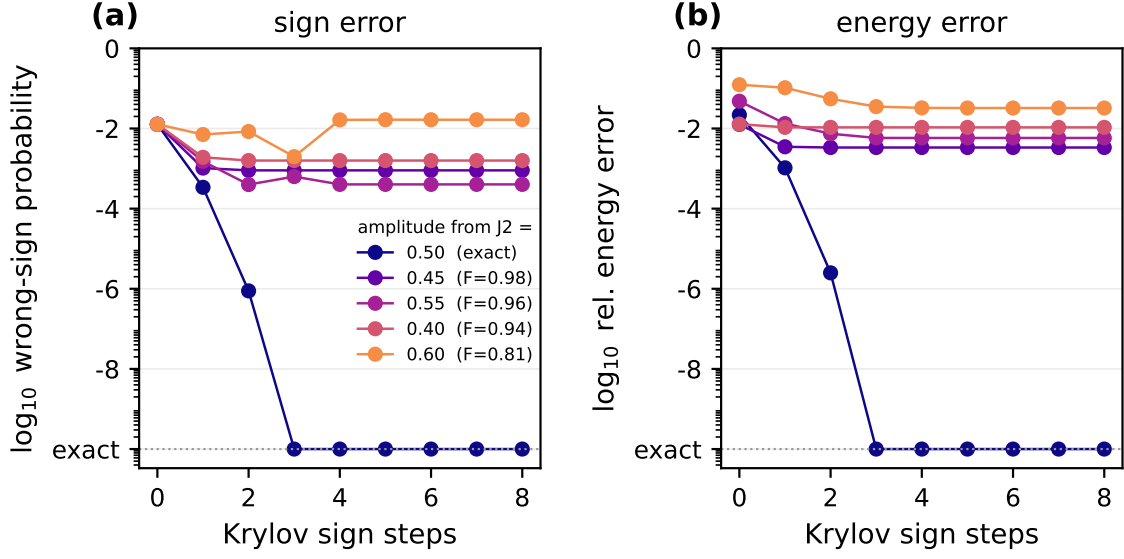


FIG. 3. Krylov sign steps on an approximate amplitude at $J_2/J_1 = 0.5$; the amplitude is the exact one of J'_2 (fidelity F). (a) Sign and (b) energy error vs number of Krylov sign steps, compared with ED.

With an approximate amplitude the steps still lower the energy, but they converge to a nonzero error set by the amplitude [Fig. 3(a,b)]. For an amplitude with fidelity 0.98 ($J'_2 = 0.45$) the energy error drops from 1.3×10^{-2} to 3.4×10^{-3} , and the wrong-sign probability from 1.3×10^{-2} to 9×10^{-4} . Only the exact amplitude reaches the exact signs; the residual error is set by the amplitude, which motivates updating the amplitude itself (Sec. III B).

3. Neural-network amplitude on 6×6

On 6×6 no exact amplitude exists. We take the positive amplitude of the optimized ViT state, give it Marshall signs, and apply one Krylov step. The ViT's own signs serve as a reference.

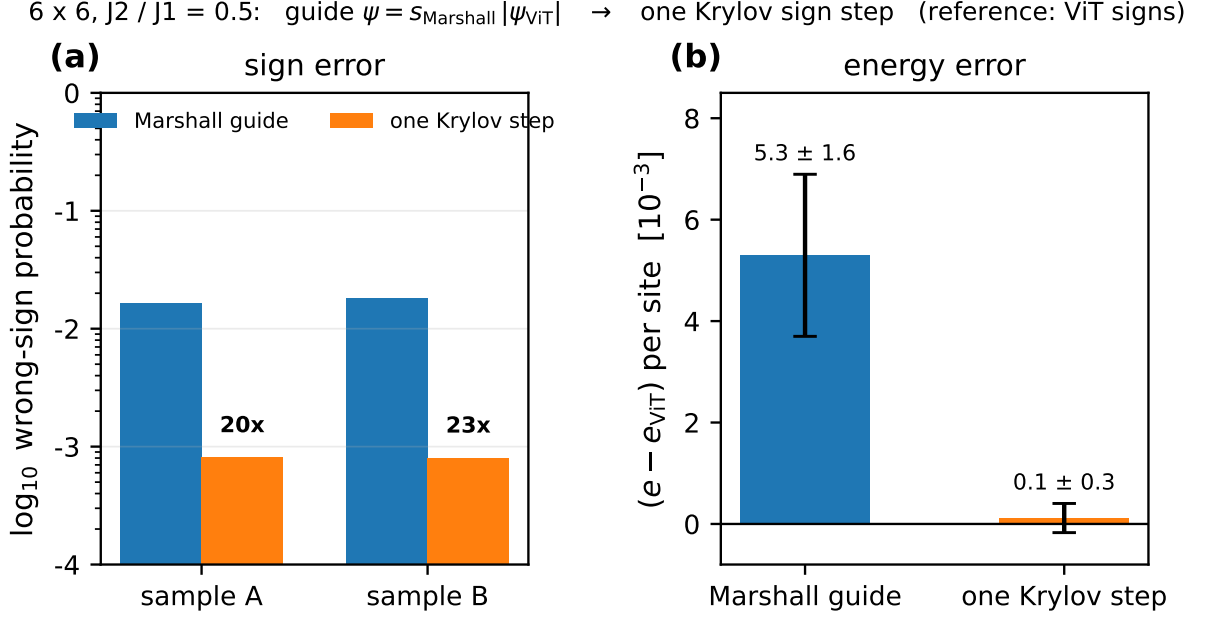


FIG. 4. One Krylov sign step on the ViT amplitude (6×6). No ED is available, so the ViT signs are the reference. (a) Sign error and (b) energy error per site of the Marshall guide and after one step.

One Krylov step reduces the disagreement with the ViT signs by a factor 20–23 [Fig. 4(a)]. With the same amplitude, Marshall signs lie $(5.3 \pm 1.6) \times 10^{-3}$ per site above the ViT signs, while the Krylov signs are statistically indistinguishable from them, $e_{\text{Krylov}} - e_{\text{ViT}} = (1.2 \pm 2.9) \times 10^{-4}$ [Fig. 4(b)]. A single label-free step thus recovers the energetically relevant sign structure of an optimized neural network.

B. Fixed-node–Krylov loop

Krylov steps give the best signs for a given amplitude; fixed node gives an improved amplitude for given signs, with $E_0 \leq E_{\text{FN}} \leq \langle H \rangle_T$ [6]. We alternate the two, here with exact diagonalization of the fixed-node Hamiltonian (no sampling):

1. **Guide.** Exact $J_2 = 0$ amplitude with Marshall signs, target $J_2/J_1 = 0.5$. Krylov steps alone do not change these signs.
2. **Fixed node.** Diagonalize H_{FN} of the current guide; its ground state ϕ_{FN} is the new amplitude.
3. **Krylov sign step** on ϕ_{FN} , Eq. (5).
4. **Repeat** from step 2.

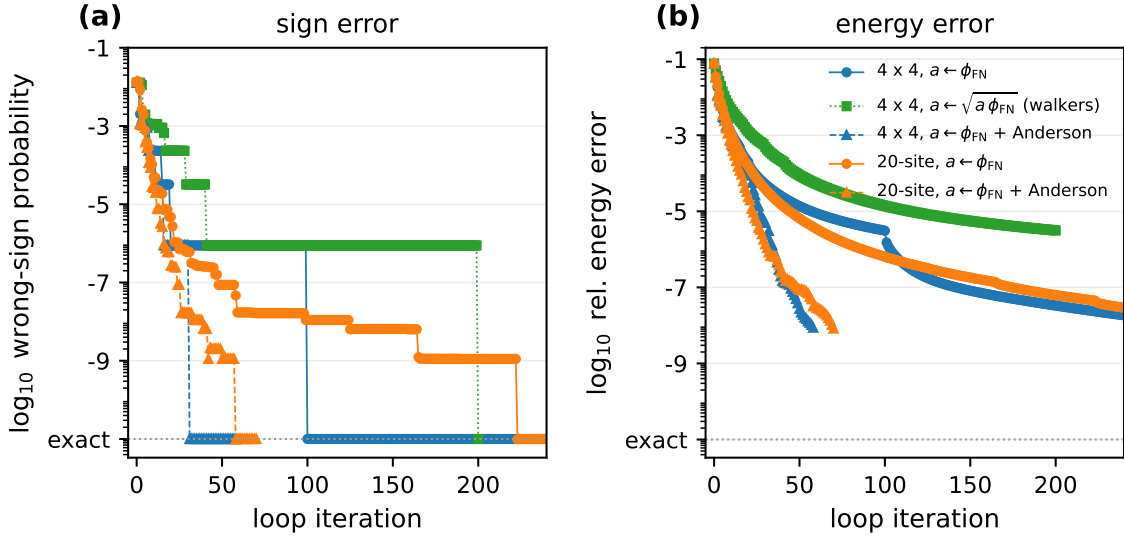


FIG. 5. Fixed-node-Krylov loop with exact diagonalization, started from the $J_2 = 0$ amplitude with Marshall signs, for three amplitude updates: $a \leftarrow \phi_{\text{FN}}$; $a \leftarrow \sqrt{a} \phi_{\text{FN}}$, as learned from walkers; and $a \leftarrow \phi_{\text{FN}}$ with Anderson extrapolation from the last three iterates. (a) Sign and (b) energy error vs iteration, compared with ED. All values are exact (no statistical error).

The loop reaches the exact ground state (Fig. 5): exact signs after 100 iterations on 4×4 and 224 on the 20-site cluster. The slowness comes from the amplitude, since the fixed-node Hamiltonian depends on the guide amplitude through the moved matrix elements. Anderson acceleration of $\log a$ cuts this to 31 and 59 iterations. The geometric-mean update $a \leftarrow \sqrt{a} \phi_{\text{FN}}$ that walker learning provides converges as well, exactly twice as slowly (200 iterations). Exact diagonalization of H_{FN} costs as much as ED itself, so this proves the idea, not its scaling.

C. Sampled fixed node on 4×4

At scale, fixed node returns walkers, not ϕ_{FN} . To isolate the effect of sampling, we repeat the loop of Sec. IIIB on 4×4 but estimate ϕ_{FN} from N samples of the walker distribution $f \propto a \phi_{\text{FN}}$: on this small cluster every configuration is visited, so $\phi_{\text{FN}}(x) \propto \text{count}(x)/a(x)$ needs no network. Everything else stays exact.

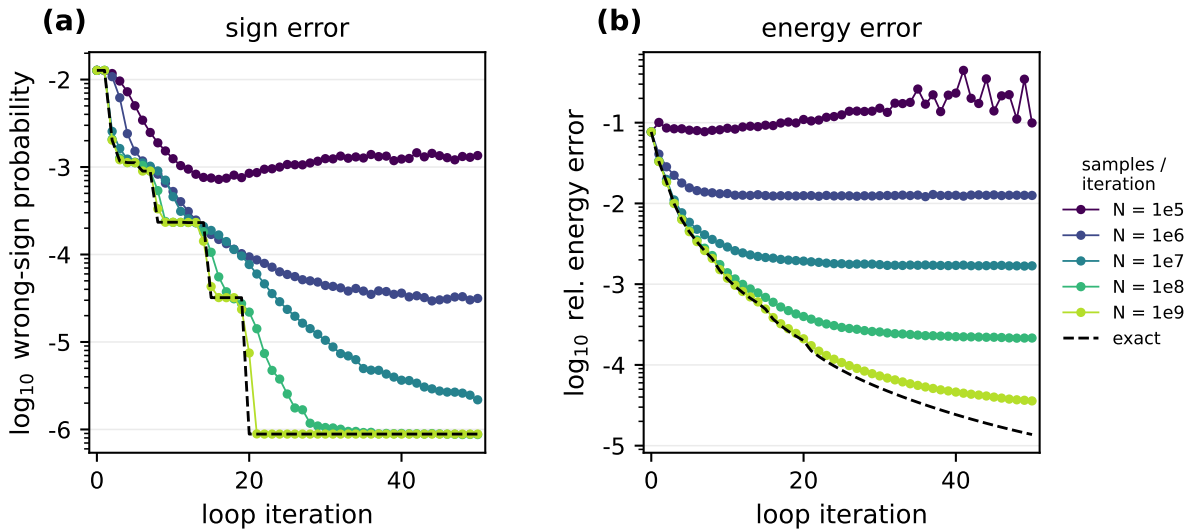


FIG. 6. Fixed-node-Krylov loop on 4×4 with ϕ_{FN} estimated from N independent samples per iteration (mean over four seeds; two for $N = 10^9$), and the exact loop. (a) Sign and (b) energy error vs iteration, compared with ED.

Sampling noise sets a floor (Fig. 6). The energy error plateaus at 1.3×10^{-2} , 1.7×10^{-3} and 2.2×10^{-4} for $N = 10^6$, 10^7 and 10^8 , so the excess over the exact loop falls roughly as $1/N$; from $N = 10^8$ the signs follow the exact loop. With $N = 10^5$ most configurations go unvisited and the loop degrades. Real fixed-node walkers are worth fewer samples than independent draws: at 10^6 and 10^7 walker snapshots the loop matches about 3×10^5 and 4×10^6 independent samples.

D. First iteration at scale: 6×6 and 8×8

One iteration needs no learned amplitude: the guide is the ViT amplitude with Krylov signs (from Marshall), and fixed node returns its energy directly from the walkers. We compare it with the ViT itself and with fixed node guided by the full ViT (its own amplitude and signs), all at $M = 128$, $\beta = 1.2$, two walker populations.

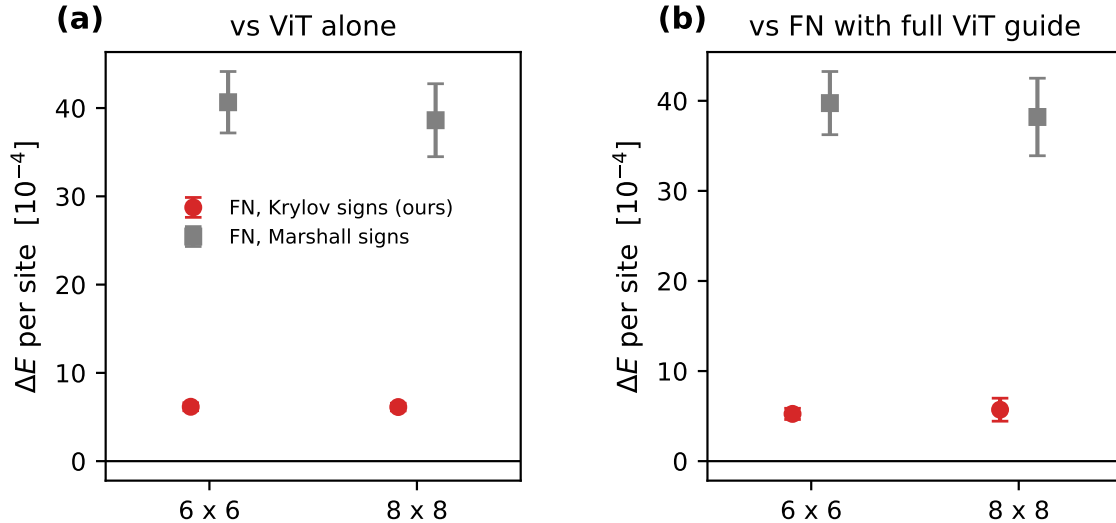


FIG. 7. One untrained iteration on 6×6 and 8×8 : fixed-node energy with Krylov or Marshall signs on the ViT amplitude, relative to (a) the ViT variational energy and (b) fixed node with the full ViT guide.

Krylov signs recover most of the gap between Marshall and the network signs, from 4×10^{-3} to 6×10^{-4} per site on both lattices, but not all of it (Fig. 7). Fixed node does not close the remaining gap: our energies, $-0.502984(18)$ on 6×6 and $-0.498210(25)$ on 8×8 , lie above the ViT itself, $-0.503600(41)$ and $-0.498823(35)$. Fixed node with the full ViT guide reproduces the ViT energy within errors, which bounds the fixed-node bias at this M to about 10^{-4} per site. Figure 8 places all values among published results.

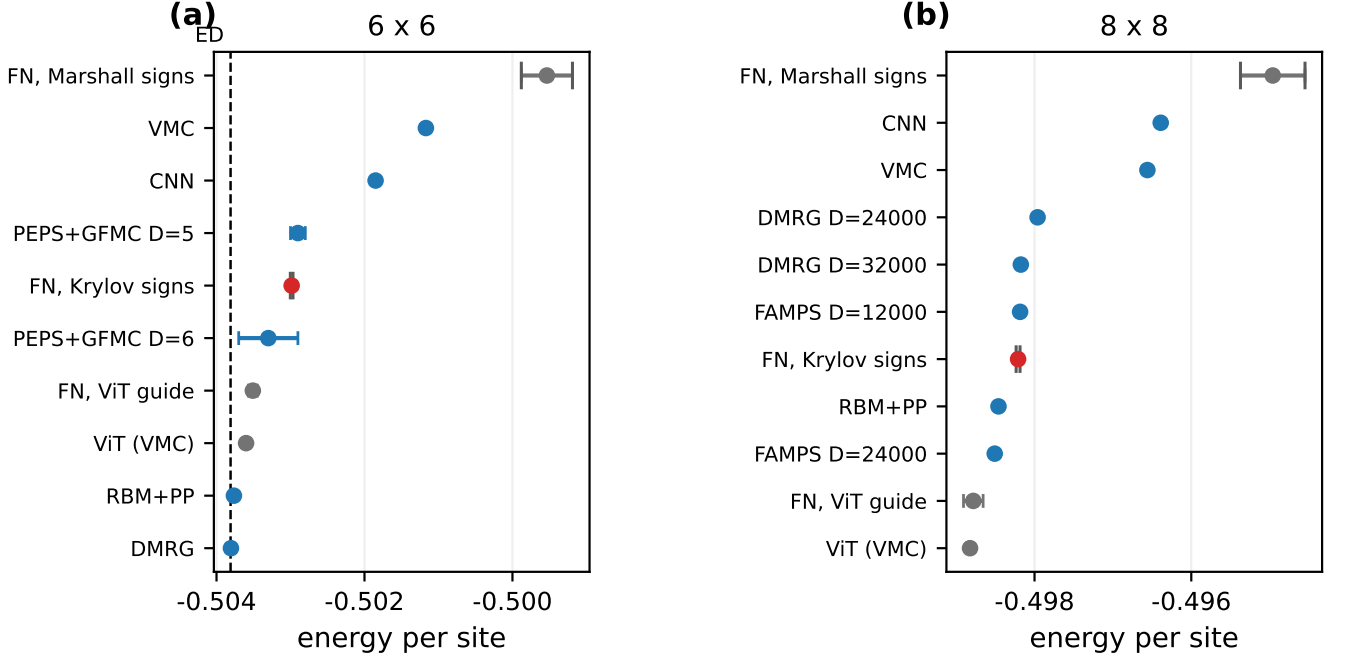


FIG. 8. Energies per site at $J_2/J_1 = 0.5$: our fixed-node values (Marshall, Krylov, full ViT guide; $M = 128$, not extrapolated), the ViT, and published results: (a) 6×6 , (b) 8×8 . ED [9], VMC [10], CNN [4], DMRG [11], RBM+PP [12], PEPS+GFMC [13], FAMPS [14].

E. Closing the loop at scale: learning the amplitude

On 6×6 , closing one loop iteration at $M = 128$ with a binned amplitude correction leaves the signs unchanged and gives a next amplitude correction statistically consistent with zero, with $E_{\text{FN}} = -18.1130(36)$ from two populations. Within this resolution the 6×6 loop has reached a fixed point.

To iterate further, the walker information must become a positive amplitude that can be evaluated on any configuration, i.e. a learned network. Walkers sample $f \propto a \phi_{\text{FN}}$, so maximum likelihood gives the geometric-mean update $a_{\text{new}}^2 \propto a \phi_{\text{FN}}$, Eq. (8). On 8×8 we train the ViT amplitude toward this target with SR, apply a Krylov step, and rerun fixed node (Fig. 9).

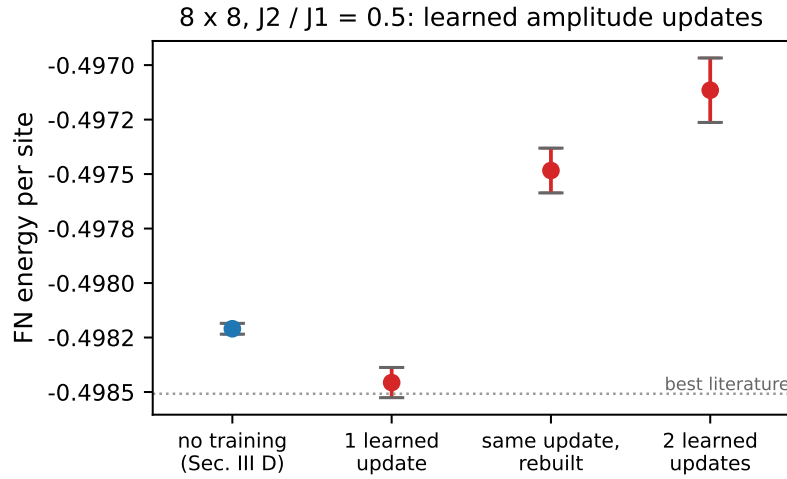


FIG. 9. 8×8 fixed-node energy per site along the learned loop (two walker populations each). Dotted: best same-geometry literature value [14].

Starting from the untrained result of Sec. III D (-0.49821 per site, above the ViT), one learned update gave -0.49846 , but an independent rebuild of the same update gave -0.49748 , and a second update -0.49711 , both above the start. Keeping the signs after the first update and changing only the amplitude already raises the energy, so the learned amplitude, not the sign step, fails. Learning a callable amplitude accurate enough for fixed node is the open problem.

IV. DISCUSSION

Splitting $\psi = sa$ separates two problems of different difficulty. The sign problem, given the amplitude, is easy: a label-free threshold on the local energy captures the frustrated sign corrections, a few repetitions reach the exact signs on exact clusters, and on 6×6 one step matches the signs of an optimized neural network energetically. With an approximate amplitude the same step returns the best signs for that amplitude, so the residual error is an amplitude error.

The amplitude problem is sign free once the signs are fixed, and lattice fixed node solves it with an upper-bound guarantee. Alternating both updates exactly reaches the ground state; the loop is slow because H_{FN} depends on the guide amplitude, and Anderson mixing speeds it up threefold. With sampled fixed node the loop still converges, but to a floor that falls roughly as the inverse number of samples; on 4×4 , 10^7 – 10^8 samples per iteration are needed to follow the exact loop, and correlated walkers are worth about three times fewer.

On 6×6 and 8×8 one untrained iteration brings the fixed-node energy within 6×10^{-4} per site of the underlying network but not below it. Iterating requires a callable amplitude learned from fixed node. For the energy to decrease with certainty, the learned guide must reach a variational energy below the previous fixed-node energy, since $E_{\text{FN}}[a_{k+1}] \leq \langle H \rangle_{a_{k+1}}$; on 8×8 our learned updates did not, and the energy rose. The consistent objective is the energy of the network under the frozen fixed-node Hamiltonian, which on exact 4×4 recovers 99.7% of the fixed-node gain; its finite-sample behaviour is the open question. Fixed node then serves as the referee of each learned step and supplies the reported energy.

The approach also requires a useful starting amplitude and differs from repeated Lanczos steps: keeping the Krylov amplitude makes the energy worse on 6×6 (Appendix A).

Appendix A: Direct second Krylov step

On 6×6 , retaining the full first Krylov amplitude gives $a_1(x) = |T_1 - r_0(x)|a_0(x)$ with coefficient of variation about 0.18. A second label-free threshold changes only 0.8–2.0% of the $|\psi_{\text{ViT}}|^2$ probability but raises the fixed-amplitude energy by 0.65–0.96. The feedback construction therefore re-solves the amplitude before another sign update.

Appendix B: Statistical conventions

Exact-diagonalization data are deterministic up to eigensolver precision. For 6×6 fixed-amplitude energies we quote between-chain standard errors from 64 chains. For fixed-node data with two independent walker populations, error bars denote half the population difference and are reproducibility scales only; the 6×6 Marshall value uses the standard error over eight populations.

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